

End-of-studies internship report

Predicting ecological partners for intestinal colonization by
Bifidobacterium adolescentis using genome-scale metabolic
models (GSMM)

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Résumé

Abstract — Abstract — We explored an explainable strategy to identify ecological partners that optimize the colonization of a target intestinal strain that is predicted to be beneficial to its carriers. To identify optimal partners, we first reconstructed genome-scale metabolic models (GSMMs) from 40 public genomes using ModelSEEDpy, complemented by *gapfilling* (objective bio1). We unified exchange reactions (EX_) and generated a cloud of 10 000 media by fixing water and controlling oxygen. We then employed static and dynamic flux balance analysis (FBA and dFBA) to investigate how different strains affect the growth of the target strain under varying environmental compositions. To capture broader conditions and to make our conclusions robust to environmental uncertainty, we generated a cloud of 10,000 different media compositions because we cannot precisely determine the environments where the strain will be enriched and delivered, including the gut environment. To reduce the computational cost while preserving the best candidates that support the growth of the target strain *Bifidobacterium adolescentis*, this strategy reduced the computational time from ~ 4 h per environment (exhaustive pre-screen) to ~ 9 min with beam search. Finally, we aggregated partner occurrence frequencies over 10 000 media and documented the cases where assembly improves or degrades the growth of the target. These results will be forwarded to wet-lab collaborators to measure real data and assess the quality of our computational prediction on real data.

Keywords : microbiome, GSMM, FBA, dFBA, beam search, gapfilling, cross-feeding.

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List of abbreviations and symbols

GSMM

Genome-Scale Metabolic Model.

FBA / dFBA

Flux Balance Analysis / dynamic FBA.

ORF

Open Reading Frame.

SBML

Systems Biology Markup Language (exchange format).

EX_

Prefix of exchange reactions (medium).

HPC / SGE

High-Performance Computing / Sun Grid Engine.

μ Optimal growth rate from an FBA.

X_i Biomass of population i .

S_j Concentration of metabolite j in the environment.

v_{ij} Net flux associated with metabolite j by population i (output of FBA).

K Monod constant (here $K = 10$).

MAX_UPTAKE

Upper bound of import (here 10).

YIELD_FACTOR

Damping of secretions (0,1 when $v > 0$).

1 Context and motivation

Improving the durable colonization of beneficial bacteria in the human gut is a central goal of microbiome biotechnology. Large-scale analyses show that the gut microbiota is organized into *global branches* and *local states* shaped by diet, environment and host-intrinsic factors, implying strong context dependence [1]. Introducing *ecological partners* enabling metabolic complementarity (cross-feeding, removal of toxic metabolites, resource sharing, etc.) thus becomes a key strategy to secure the implantation of a target strain.

Genome-scale metabolic models (GSMs) and flux balance analysis (FBA/dFBA) offer a quantitative and explainable route : starting from genomes, we reconstruct networks, impose environmental constraints (medium, electron acceptors) and predict growth and exchanges of a strain alone or in community [3]. The operational objective is to turn the question “*which partners favor colonization of the target strain ?*” into a constrained optimization problem, testable *in silico* and then experimentally validated.

At the end of simulations on 10 000 environments, the most promising combinations will be passed to our *wet-lab* partners to **collect in vitro / in vivo data**. These data will make it possible to **train and validate the final predictive model**, a necessary condition to close the design-and-evaluation loop.

2 State of the art

2.1 Inter-individual variability and contextual design

Massive cohorts reveal latent architectures of the gut microbiota (*global branches*, *local states*) associated with distinct metabolic regimes [1]. This

motivates a *contextual* design of microbial assemblies (target strain + partners) rather than a single recipe. *In practice, suitable partners may also enhance the target’s growth within fermentation matrices prior to delivery, increasing the likelihood of long-term colonization once ingested.*

2.2 From sequence to model : annotation, reconstruction and *gapfilling*

A **genome-scale metabolic model (GSMM)** is a structured representation of an organism’s metabolic network reconstructed from its genome : genes $\rightarrow$ enzymes $\rightarrow$ reactions assembled into a stoichiometric network. This reconstruction enables simulations under defined media to predict growth and metabolite exchanges in monoculture or community settings.

Resources such as **ModelSEED** [7] and strain-specific databases (**AGORA2** [5]) structure GSMM reconstructions ; modern pipelines (**gapseq**, **Bactabolyze**) improve their quality [9, 10]. AI-guided *gapfilling* (**iScience 2024** [2]) and other ML approaches [11] fill in missing reactions.

2.3 Intestinal media and normalization of exchanges

Before a new strain or its partners can be introduced into the gut, they must be grown in appropriate culture media ; for delivery through fermented foods, adaptation to the *fermentation matrix* is also needed. Accurately specifying these media and the gut environment is challenging due to inter-individual variability.

In the gut, anaerobiosis dominates ; dissolved oxygen is limited. Modeling therefore proscribes O_2 as electron acceptor, except in controlled scenarios [3]. *Normalization of exchange reactions* (**EX...**) and a *unified list* of exchanges allow application of a common medium and sampling of a *cloud of media* (Dirichlet draws).

2.4 Community modeling and stationary pre-screen

Community frameworks such as **MICOM** [6] rely on FBA. Scoring tools (**SMETANA**) serve as a *pre-screen* before more expensive analyses [14, 15, 16]. Recent evaluations confirm the value of FBA-based approaches for predicting interactions [12].

Dynamic FBA (dFBA) extends FBA by coupling flux optimization with time-evolving extracellular conditions, iteratively updating fluxes and concentrations to capture substrate depletion, product accumulation and delayed cross-feeding. **Target strain.** Our study focuses on *Bifidobacterium adolescentis*, a common gut commensal linked to beneficial host effects (e.g., immune modulation, metabolic balance). Establishing this species in adult microbiomes can be difficult due to ecosystem stability and competition; *identifying supportive partners* that supply nutrients, remove inhibitors or facilitate adaptation (e.g., within fermented food matrices) may increase colonization success.

3 Materials and Methods

3.1 Data (40 public genomes)

We use a **set of 40 bacterial genomes** of food and intestinal origin, **from public repositories** (e.g., NCBI/ENA). Source files are provided in `.fna` (nucleotide) and/or `.faa` (protein), possibly compressed as `.gz`. They are versioned in the working repository (`genomes/`) and the reconstructed models are archived in `models_gapfilled/`. **No private genome is used.** Metadata (species, isolation source, accession) and the file manifest are listed in the appendix.

Practical relevance of the panel. Several strains originate from plant-based fermentation matrices, making them realistic and safe candidates for delivery via fermented foods. Including both food-associated and intestinal strains increases the chance of finding *ecological bridges* that survive in the food matrix and later in the gut, potentially supporting the growth of *B. adolescentis*.

Consistency checks (I/O). Before reconstruction, we verify integrity (non-zero size, readable FASTA), uniqueness of sequence identifiers and encoding compatibility. `.faa.gz` files are decompressed; `.fna` are converted to `.faa` (see §3.3.1).

3.2 Tools and software (code excerpts)

- **Language** : Python (3.x).
- **Metabolic modeling** : COBRApy (SBML I/O, optimization), modelseedpy (draft construction).
- **LP solvers** : **GLPK** (default for dFBA), **Gurobi** (spot validation and fast tests).
- **Annotation / ORF** : pyrodigal (ORF prediction from `.fna`, metagenomic mode), Biopython (sequence I/O).
- **Gapfilling** : dnngior (deep learning), objective bio1.
- **Scientific stack** : NumPy, Pandas, SciPy (solve_ivp), Matplotlib.
- **HPC** : SGE (Sun Grid Engine) for scheduling (`qsub`, array jobs), `bash` for submission; `conda` environment management.
- **Versioning** : Git/GitHub (`git clone` of `genomes` and `models_gapfilled` repositories).

3.3 Algorithmic and computational framework (step-by-step details)

Additional code repository : <https://github.com/Shinzen31/content.git>

3.3.1 Preprocessing and rationale for choices

Why predict ORFs (Pyrodigal) ? Some public inputs are in `.fna` (genomes or contigs). To guarantee a unified pipeline (protein-based), we use **Pyrodigal** in metagenomic mode to detect potential genes and generate a coherent `.faa` per genome. This choice :

- avoids biases due to heterogeneous original annotations ;
- provides a *standardized inventory of proteins* for the ModelSEED phase ;
- remains robust to fragmented assemblies (the *meta* mode).

ORFs are written in FASTA (`.faa`). A minimal example is provided in the code : opening `.fna.gz`, iterating sequences with **Biopython**, prediction via `pyrodigal.GeneFinder(meta=True)`, outputting translations.

3.3.2 Draft GSMM construction

Each proteome (`.faa`) is passed to `modelseedpy:MSGenome.from_fasta` creates a genomic representation, then `MSBuilder.build_metabolic_model` produces a draft :

- **Default compartmentation** : cytosol `c0`, extracellular `e0` ;
- **Standardized biomass reaction** : `bio1` set as objective ;
- **Export** to SBML (`.xml`) for compatibility and traceability.

Why start from an automated ModelSEED draft ? To reduce manual biases, ensure inter-strain homogeneity and benefit from a heavily curated reaction base [7]. Strain-specific reconstructions (AGORA2) [5] are used as quality references and to guide some corrections.

3.3.3 Anaerobic reference medium and O₂ filtering

The gut is **mostly anaerobic**. We therefore build an *oxygen-free* reference medium (`complete_media_without_oxygen.tsv`) by **excluding** any metabolite containing oxygen-related keywords (`o2`, `oxygen`, `superoxide`, `peroxide`, etc.). To avoid false positives (e.g., `co2`), filtering uses **word boundaries** (regex `\b`). **Why ?** Forcing viability under anaerobic constraints helps reveal the *true electron acceptors* and fermentative pathways relevant to the colon [3].

3.3.4 Why gapfilling, and how ?

Incomplete genomes, missing transporters or partial annotations produce *non-viable drafts* under realistic constraints. **Gapfilling** aims to *minimally*

complete the network to restore growth (**bio1**).

- **Preparation.** We impose the *anaerobic medium* defined in §3.3.3; we ensure that extracellular **EX_** exist for all relevant metabolites (*function fix_exchange_reactions*).
- **Algorithm.** We use **dnngior** (deep learning) to propose minimal additions, with `objectiveName='bio1'`. This approach is motivated by recent work showing the benefit of AI to complete GSMs of varying incompleteness [2, 11].
- **Post-processing.** After gapfilling : (i) force **bio1** as objective; (ii) rewrite to SBML; (iii) *sanity check* : simple FBA optimization and inspection of **EX_** exchanges.

Why this strategy ? It reconciles *realism of the medium* (anaerobic), *parsimony of additions*, and *reproducibility* (same settings for all strains).

3.3.5 Unification of exchange reactions

For each “gap-filled” model we extract **EX_** reactions having `lower_bound < 0` (thus import possible) and build a **unique catalog** `group_medium_exchanges.tsv` :

- **Why unify ?** To test identical common media for all models, avoid nomenclature divergence and facilitate coherent random sampling.
- **Implementation.** Deduplication by reaction identifier ; keeping metabolite identifier, name and formula ; tabular writing.

In practice, for each gap-filled model we also collect all metabolites that can be exchanged and ensure the corresponding **EX_** reactions exist before consolidating them into this unique catalog.

3.3.6 Generation of the *environment ball* and justification

From the unified catalog, we sample **relative concentrations** via a **Dirichlet** law $\text{Dir}(\alpha = 1)$. **Reasons :**

- **Non-informativeness** (all components a priori symmetric) $\Rightarrow$ “neutral” exploration of a large nutritional space ;

- **Sum invariance** $\Rightarrow$ vectors represent *relative budgets*, easily rescalable ;
- **Control** of some key metabolites : **water** [EX_cpd00001_e0] is **fixed** at 10, which stabilizes optimization (avoids a “dry” medium) and reflects its quasi-constant availability.

Implementation. Media are recorded in `environment_ball.tsv` with a column `env_id` (up to **10 000** environments). A sensitivity variant adds **fixed oxygen** ([EX_cpd00007_e0]=100) to test microaerophily, without using it in the main optimization.

3.3.7 Stationary pre-selection (FBA) and *beam search*

For a **target strain** and 39 potential partners, we consider groups of total size $g \in \{2, 3, 4, 5\}$ (target + $g - 1$ partners). The number of **unordered combinations** per environment is

$$\sum_{k=1}^4 \binom{39}{k} = 39 + 741 + 9\,139 + 82\,251 = \mathbf{92\,170}.$$

Why pre-select with static FBA ? Because an “all-dFBA” would be prohibitive : 2–3 min *per group*. The **stationary pre-screen** (FBA/SMETANA) is common practice and *preserves the ranking of the best assemblies* well, despite partially losing higher-order interactions [14, 15, 12].

Score and *beam search*. We define a **group score** as the ratio $\mu_{\text{group}}/\mu_{\text{solo}}$ of the target under the current medium. *Beam search* (width 10) iteratively explores the partner space by retaining only the best candidates at each depth ($g - 1$ partners to choose). **Measured costs** : ~ 4 h/environment for exhaustive static vs ~ 9 min/environment with beam.

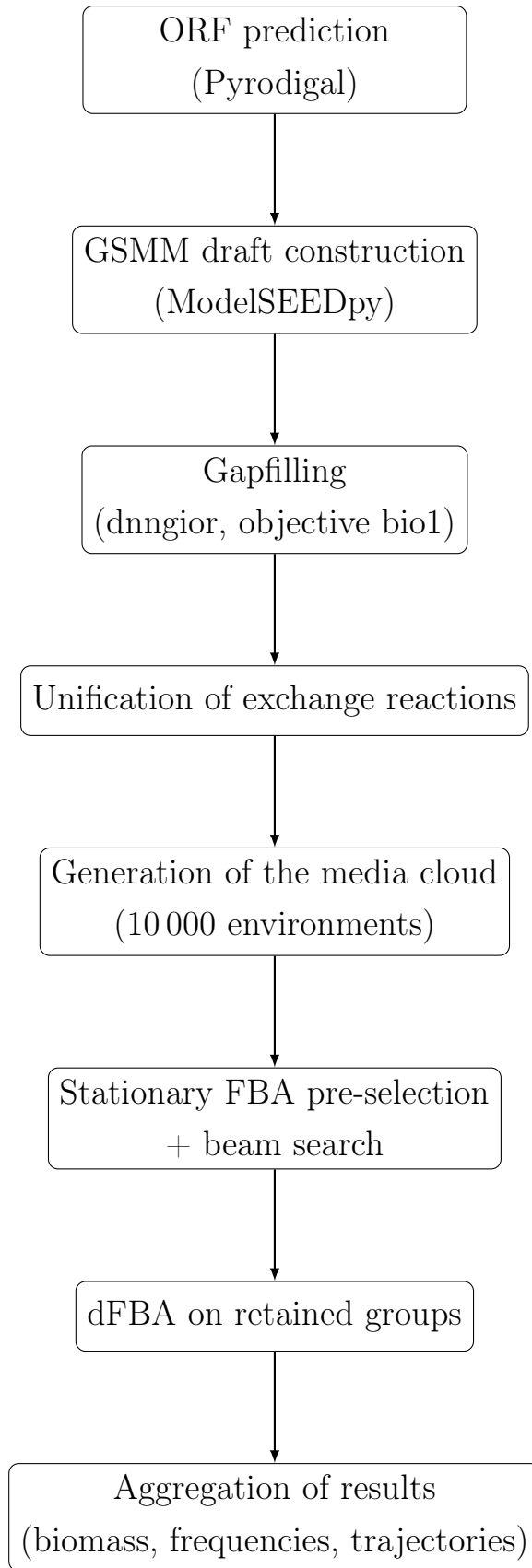


FIGURE 1 – Workflow diagram : steps from reconstruction to aggregation of results.

Precision note. Stationary methods (FBA, SMETANA, community frameworks) generally *preserve* the top ranks [14, 15, 12]; they can ignore some multi-species synergies/antagonisms revealed *in dynamics*, but the impact on *pre-screening* is limited in practice.

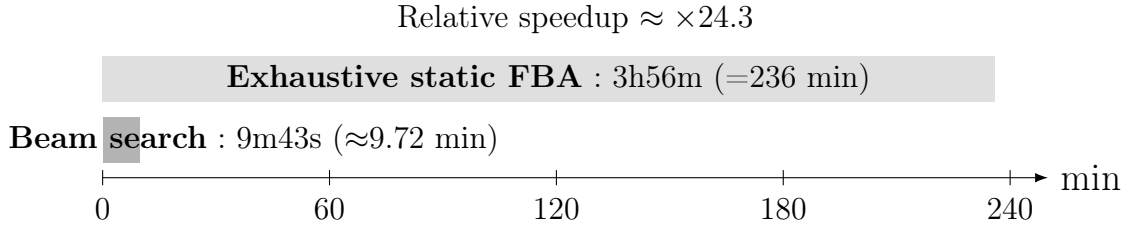


FIGURE 2 – Validation on a reference environment : *exhaustive static FBA* (3h56m) vs *beam search* (9m43s) to identify the best group ; comparison of time cost.

3.3.8 Why a dFBA, and how ? (consistent with dFBA.py)

Motivation. Stationary FBA neglects **substrate depletion**, **product accumulation**, delays in *cross-feeding* appearance and **extinctions**. **dFBA** couples an ODE system with successive FBAs to reproduce these transient effects [13, 12].

Formulation. For each population i (biomass X_i) and environmental metabolite j (concentration S_j) :

$$\frac{dX_i}{dt} = \mu_i(S) X_i, \quad \frac{dS_j}{dt} = \sum_i X_i v_{ij}(S),$$

where μ_i and v_{ij} come from an FBA with **import bounds** of **Monod** type :

$$u_r(S_r) = \frac{S_r}{S_r + K}, \quad K = 10, \quad \text{lower_bound} = -u_r, \quad \text{upper_bound} = \text{MAX_UPTAKE} =$$

Numerical stabilizers : projection $S_j \geq 0$ (`avoid_zero_met`) ; **damped secretions** ($v > 0 \Rightarrow 0.1v$) with `YIELD_FACTOR=0.1` ; **water** enforced via `water [EX_cpd00001_e0]  $\geq 10$` ; **irreversible extinction** if $X_i < 10^{-8}$.

Integration settings. Horizon $[0, 48]$ h, $N_points = 40$, solver **GLPK**. Initial conditions : $X_i(0) = 0,1$ for each member, $S_j(0)$ from the current medium.

Cost. **2–3 min** per group. Hence the interest of the **pre-screen** (§3.3.7) before dFBA.

3.3.9 Why not an “all-dFBA” ? (order of magnitude)

Without pre-selection, simulating *all* 92 170 combinations per environment in dFBA would cost :

$$92\,170 \times (2-3) \text{ min} = 184\,340-276\,510 \text{ min} = 3\,072-4\,608 \text{ h} = 128-192 \text{ CPU day}$$

At the scale of **10 000 media** : **30,72–46,08 million CPU hours** ($\sim 3\,506-5\,260$ years). Hence the strategy **beam + static FBA** $\Rightarrow$ **9 min/env** instead of 4 h in pre-screening.

3.3.10 Parallel orchestration and reproducibility

Orchestration. `run.py` orchestrates by environment *chunks* : for each `env_id` and each group size $g \in \{2, 3, 4, 5\}$, it launches `dfba.py` with `timeout` (600–1 200 s), progressively merges outputs (`results_bs{g}/dfba_final_biomass.csv`) and saves trajectories (`trajectories/*`). A summary `partner_frequency.csv` (partner occurrence frequency) is generated.

HPC (SGE). The `qsub` script submits an *array* of 100 tasks, each processing 100 environments (0–9 999 covered). Typical configuration : `-pe thread 16` (4 group sizes $\times$ 4 env in parallel), `h_vmem=4G/slot`, `h_rt=12:00:00`. To limit memory fragmentation : `MALLOC_TRIM_THRESHOLD=1`. The `conda` environment is activated at the start and closed at the end of the task.

Reproducibility. Pseudo-random seed `random_state` fixed when generating media ; SBML versioned ; centralized numerical parameters ($K = 10$, `MAX_UPTAKE=10`, `YIELD_FACTOR=0.1`, `EXTINCT_EPS=10-8`, `water = 10`). CSV *snapshots* (biomasses, metabolites, deltas) allow re-reading/post hoc analysis of trajectories.

4 Results

4.1 Time trajectories (biomass and metabolites)

dFBA enables simultaneous tracking of **growth (biomass)** and the evolution of **metabolite concentrations** in the presence of interactions (competition, cross-feeding). Figure 3 illustrates, for a given strain and a reference environment, the joint biomass/metabolite dynamics.

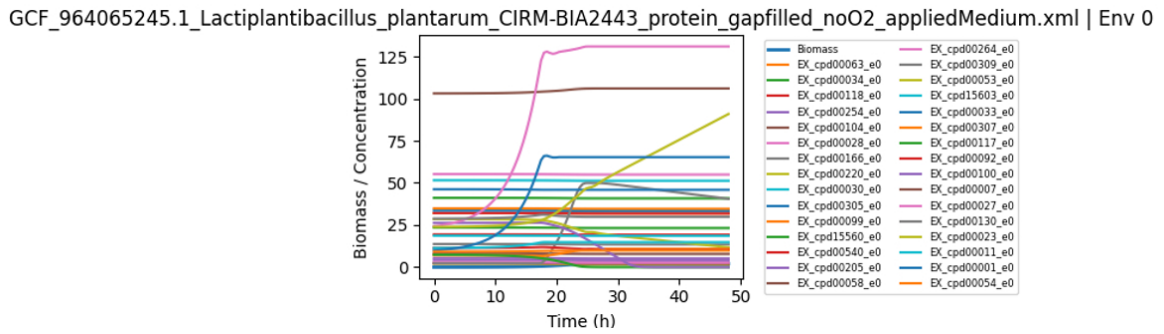


FIGURE 3 – Example of dFBA trajectories (biomass + metabolites) on a reference environment.

4.2 Occurrence frequencies in the best groups (10 000 media)

We build, for each strain and each group size (2, 3, 4, 5), an **occurrence frequency** in the best group identified per environment. Since there

are 4 different sizes, normalized frequencies range in $\{0, 0,25, 0,5, 0,75, 1\}$. The horizontal axis corresponds to strains, the vertical axis to environment identifiers.

	GCA_9640	GCA_9640	GCA_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640	GCF_9640
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0,25
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0,5
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0,5
3	0	0	0	0	0	0	0	0,25	0,25	0,25	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0,25
5	0	0,25	0	0	0	0	0	0	0	0	0	0	0	0,5
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0,25
7	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0,5
9	0	0	0	0	0	0	0	0	0	0	0	0	0	0,25
10	0	0	0	0,25	0	0	0	0	0	0	0	0	0	0,25
11	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0,25	0	0	0
13	0	0	0	0	0	0	0	0	0	0	0	0	0	0,75
14	0	0	0	0	0	0	0	0	0	0	0	0	0	0,75
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0,25
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0,25
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0,25
18	0	0,25	0	0	0	0	0	0	0	0	0	0	0	0,5
19	0	0,25	0	0	0	0	0	0	0	0	0	0	0	0,5
20	0	0	0	0	0	0	0	0	0	0	0	0	0	0,75

FIGURE 4 – Heatmap of strain occurrence frequencies in the best groups (10 000 media). Extract from the full table (the full table is too large to include here).

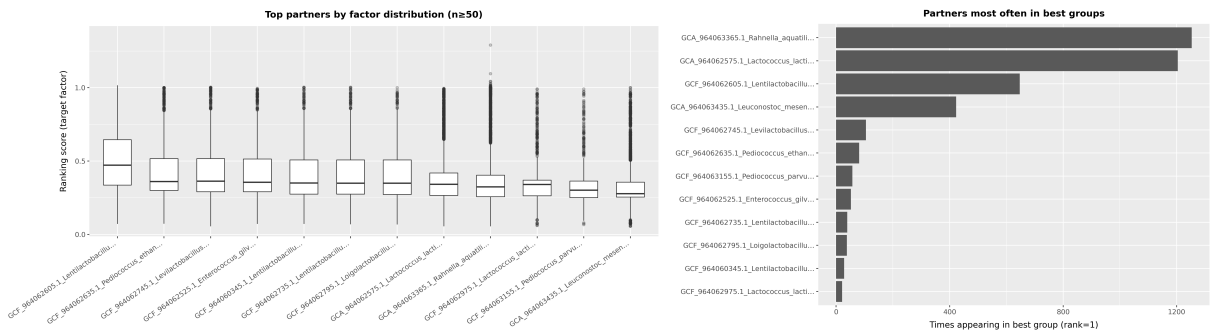


FIGURE 5 – **Relative ranking of communities.** *Left* : distribution of the *factor* (final biomass of the target in community / monoculture) per partner, $n \geq 50$; dashed line = 1. *Right* : number of times the partner appears in the *best group* (rank = 1) among $g = 2-5$ and 10 000 media. The left panel reflects gain amplitude, the right panel robustness of appearance.

4.2.1 Detailed reading of Figure 5

Overall message. The figure synthesizes, over 10 000 media and group sizes $g = 2-5$, two complementary pieces of information for each potential partner of the target. The *factor* is defined as

$$\text{factor} = \frac{\text{final biomass of the target in community}}{\text{final biomass of the target in monoculture}}.$$

The dashed line at 1 marks indifference “*community = monoculture*”. We observe that **the median is < 1 for most partners**, indicating that on average **competition dominates** (the community often penalizes the target). However, several partners show **upper tails** (points/values > 1), revealing *contextual gains* in some media.

Left panel — effect amplitude (distribution of the *factor*). Each box (partners with $n \geq 50$ observations) describes the variability of the *factor* across media and group sizes : (i) a **low and tight** box (< 1) reflects a regularly *competitive* partner ; (ii) a **higher** box and/or with **values above 1** suggests an *opportunistic* partner capable of providing a gain in specific conditions (cross-feeding, detoxification, import/export complementarity).

Right panel — robustness of appearance (count of “best groups”). Bars indicate how many times a partner belongs to the **best group (rank=1)** among $g = 2-5$ across all media. A **strong heterogeneity** is apparent : *a few partners clearly stand out* (longest bars), followed by a small intermediate core and then a *long tail* of partners rarely selected. Importantly, this count reflects **robustness of selection** under environmental constraint, *not* necessarily a gain > 1 (the best group of a medium may still be < 1 if monoculture dominates overall).

Cross-reading — practical typology for screening. Crossing the two panels allows experimental prioritization :

- **Robust generalists** : high appearance frequency (right) *and* distribution shifted upward with cases > 1 (left). *Priority candidates* for broad tests.

- **Contextual opportunists** : modest frequency but *multiple* realizations > 1 . To be tested on “favorable” media identified by nutrient analysis (§4.3).
- **Systematic competitors** : low frequency and distribution clearly < 1 . To *deprioritize* unless specific interest (e.g., growth control).

Takeaway. The figure confirms that (i) **competition is the norm** across random media (medians < 1), (ii) **a few partners** are *consistently selected* as members of the best groups, and (iii) **gains > 1 are condition-dependent**. The *FBA-pretri* $\rightarrow$ *dFBA-validation* strategy thus remains relevant : first select robust partners (right panel), then verify dynamically and *per medium* the existence of a true gain (left panel).

4.3 Medium parameters and nutrient consumption

Motivation. Beyond partner frequencies, we examine the link “**initial medium** $\rightarrow$ **outcome**” and what is **actually consumed/produced** during dFBA. This reading allows interpretation of *competition* (co-consumption of a limiting resource) and *cooperation* (production of a metabolite useful to the target).

Notation. Each metabolite is noted “**Name [ID]**”, e.g., *glucose [EX_cpd00027_e0]*. Names come from the unified exchange catalog (`group_medium_exchanges.tsv`). In the figures, labels of the form EX_cpdXXXX_e0 follow this convention.

(i) **Most consumed nutrients by the target (detailed reading of Fig. 6).** For each environment, we compute the net change $\Delta S = S_{\text{final}} - S_{\text{initial}}$ of environmental metabolites (negative values = *consumption*). Aggregating over 10 000 media, Figure 6 shows the **20 compounds with the highest mean consumption**. Three families dominate :

- **Fermentable substrates** (mainly sugars and some organic acids) — they carry most of the assimilated carbon ; co-consumption by partners

induces **direct competition** and tends to degrade the target's performance.

- **Basic mineral nutrients** (inorganic nitrogen, phosphate/sulfate, cations K^+ , Na^+ , Mg^{2+} , etc.) and **vitamins/cofactors** — their average depletion reflects sustained anabolic demands; supply by a partner can favor the target (cooperation).
- **Oxygen** [EX_cpd00007_e0] — may appear in some media when the import bound is not strictly zero; our main protocol remains anaerobic.

The 20 identifiers of the figure and their functional reading :

- **ammonium** [EX_cpd00013_e0] — **inorganic nitrogen** (NH_4^+); consumption consistent with ammonium assimilation.
- **acetate** [EX_cpd00053_e0] — organic acid that can serve as a carbon source for the target (*catalog name*).
- **EX_cpd00276_e0** [EX_cpd00276_e0] — fermentable substrate / central intermediate; recurrent consumption suggests routing to energy metabolism (*catalog name*).
- **EX_cpd00309_e0** [EX_cpd00309_e0] — ion / cofactor; participation in ionic homeostasis or coupled transport (*catalog name*).
- **EX_cpd00064_e0** [EX_cpd00064_e0] — major ion (e.g., K^+ / Na^+ depending on annotation); osmotic/transport role (*catalog name*).
- **EX_cpd00179_e0** [EX_cpd00179_e0] — vitamin/cofactor or organic acid; consumption consistent with co-enzymatic needs (*catalog name*).
- **EX_cpd00065_e0** [EX_cpd00065_e0] — major ion; electrochemical maintenance and co-transport (*catalog name*).
- **EX_cpd00322_e0** [EX_cpd00322_e0] — fermentable substrate (hexose/pentose or derivative); direct carbon source (*catalog name*).
- **EX_cpd01329_e0** [EX_cpd01329_e0] — cofactor / specialized metabolite; sensitive to biosynthetic needs (*catalog name*).
- **oxygen** [EX_cpd00007_e0] — possible traces in some media (microaerobiosis), not systematic.

- **inorganic phosphate [EX_cpd00054_e0]** — mineral anion (Pi or related); high anabolic demand.
- **EX_cpd10516_e0 [EX_cpd10516_e0]** — specific vitamin/cofactor; consumption compatible with activity of particular pathways (*catalog name*).
- **EX_cpd00209_e0 [EX_cpd00209_e0]** — organic acid / mineral salt; role in anaplerosis or ionic balance (*catalog name*).
- **EX_cpd01080_e0 [EX_cpd01080_e0]** — specialized metabolite / cofactor; marker of contextual needs (*catalog name*).
- **EX_cpd01262_e0 [EX_cpd01262_e0]** — vitamin/cofactor; potential limitation relieved by community supply (*catalog name*).
- **EX_cpd00158_e0 [EX_cpd00158_e0]** — divalent ion / enzymatic cofactor; ribosomal stabilization/enzymatic activity (*catalog name*).
- **EX_cpd00105_e0 [EX_cpd00105_e0]** — organic acid / precursor; use as energy/carbon source (*catalog name*).
- **EX_cpd00609_e0 [EX_cpd00609_e0]** — cofactor / organic nitrogen compound; signature of biosynthetic requirements (*catalog name*).
- **EX_cpd00222_e0 [EX_cpd00222_e0]** — fermentable substrate / sugar; strong contribution to assimilated carbon (*catalog name*).

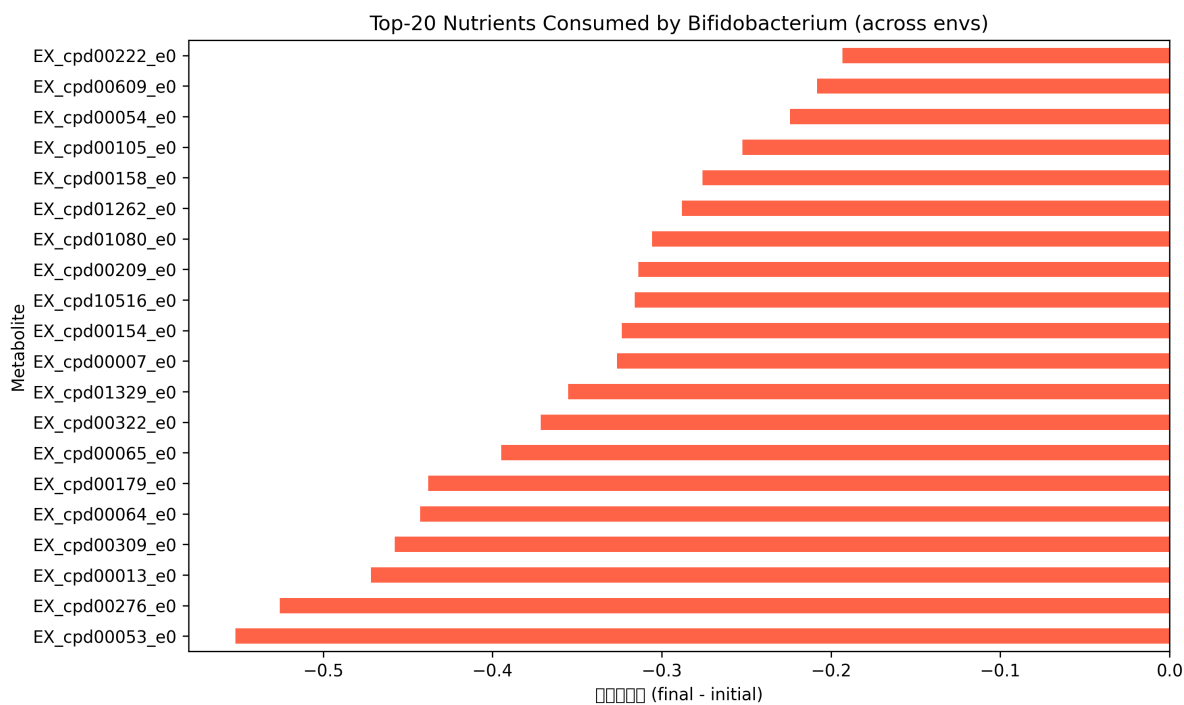


FIGURE 6 – **Top-20 nutrients consumed by the target strain** (mean ΔS over 10 000 media; negative values = consumption). Each bar corresponds to an EX_cpd..._e0 identifier (interpreted as *Name [ID]*). We distinguish : (i) fermentable substrates (carbon); (ii) mineral nutrients (N, P/S, cations); (iii) vitamins/cofactors. Co-consumption by partners induces **competition**, while supply of useful products (cofactors, reusable organic acids) supports **cooperation**.

Integrative signal (categories). The distribution of the 20 compounds among sugars, organic acids, inorganic nitrogen, mineral anions and cofactors is consistent with the physiology of *Bifidobacterium* : preference for fermentable substrates and *sustained demand* for N, P/S and coenzymes. This sheds light on the frequent observation “*monoculture > assembly*” : *many communities simply induce competition for nutrients* [4].

(ii) Map “media $\rightarrow$ outcome” (reading of Fig. 7). The PCA of media (rows = environments, columns = initial metabolites) shows a quasi-isotropic cloud (Dirichlet draws). Most points show a **low growth factor** (cool tones),

while **favorable islands** (warm tones) emerge for particular compositions.

- **Interpretation.** The *islands* likely correspond to vectors richer in **fermentable sugars** and/or **non-limiting cofactors**, reducing competitive pressure and facilitating useful cross-feeding (recycling of organic acids, factor supply).
- **Practical consequence.** A *slight rebalancing* of a few key components (assimilable carbon, nitrogen, phosphate, cofactors) can move a medium from a “cold” to a “hot” zone. These **discriminant nutritional axes** will guide medium optimization to *induce* cooperation.

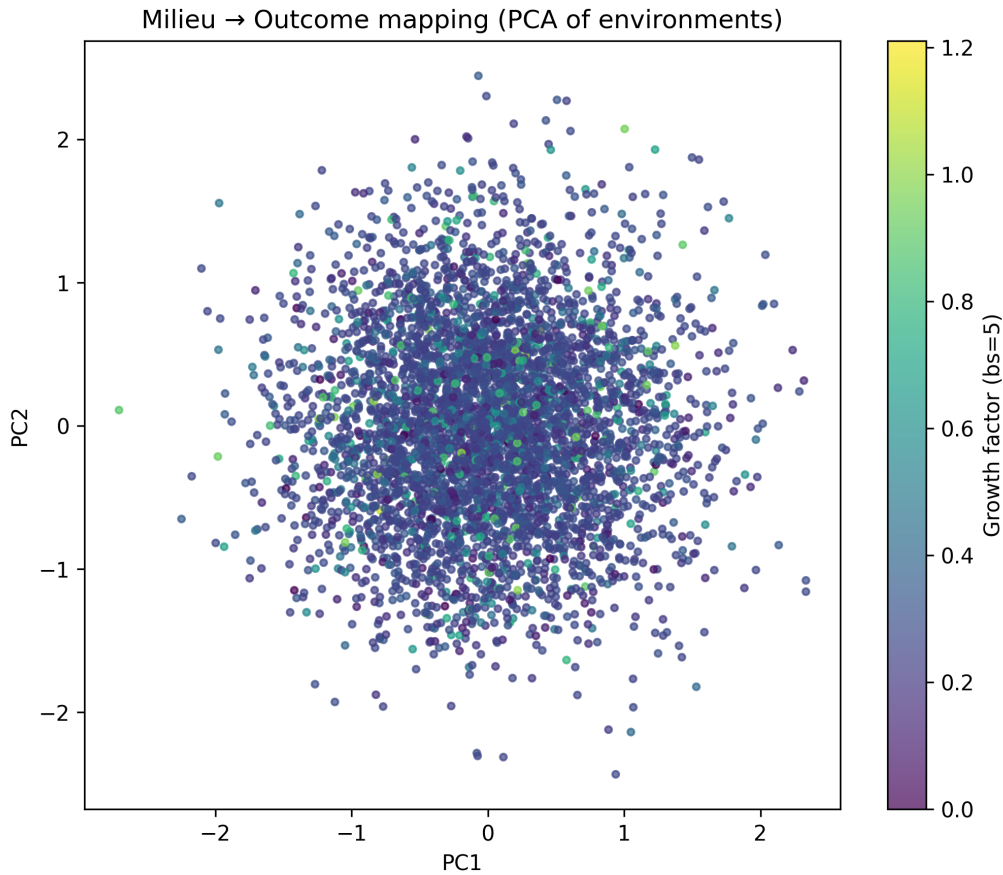


FIGURE 7 – **PCA mapping of media** (each point = an initial environment; color = target *growth factor*, bs=5). A predominantly unfavorable background (cool tones) reflects competition on key resources; *favorable islands* (warm tones) emerge when the initial composition enriches fermentable substrates and/or lifts cofactor limitations, conditions favorable to *cross-feeding*.

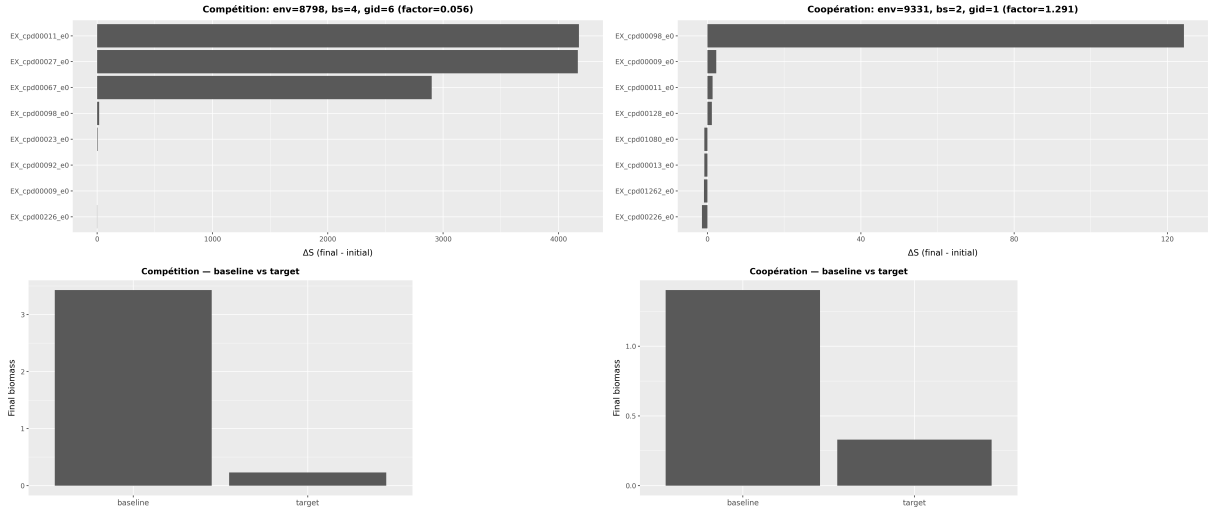


FIGURE 8 – **Concrete examples of *competition* (left) and *cooperation* (right).** *Top* : Top- ΔS of environmental metabolites ($\Delta S = \text{final} - \text{initial}$; negative = consumption; positive = production). *Bottom* : final biomass of the target — *baseline* (monoculture) vs *community*. Co-consumption of a limiting substrate leads to factor < 1 (competition); production of a metabolite useful to the target (cross-feeding/removal) leads to factor > 1 (cooperation).

4.3.1 Detailed reading of Figure 8

What is plotted. The top row displays, for a given environment, the **environmental metabolites** with the **largest net changes** during dFBA, ordered by $\Delta S = S_{\text{final}} - S_{\text{initial}}$ (negative = *consumption*, positive = *production/accumulation*). The bottom row compares the **final biomass of the target** in *monoculture* (*baseline*) and in *community* (*target*); the *factor* is $\text{biomass}_{\text{community}} / \text{biomass}_{\text{mono}}$.

Case of *competition* (left panel). Example : env = 8798, bs = 4, gid = 6, **factor** = 0.056. The final biomass of the target in community is $\sim 18\times$ *lower* than in monoculture, a sign of **strong competitive pressure**. In the top- ΔS , a few identifiers dominate (**carbon dioxide** [EX_cpd00011_e0], **glucose** [EX_cpd00027_e0], **proton** [EX_cpd00067_e0], ...) indicating **marked fluxes** on central resources. Depending on the sign of ΔS , one

reads either **co-consumption of a limiting substrate** (negative bars) or **accumulation of unvalorized by-products** (positive bars) that can penalize the target (acidification, carbon diversion, etc.).

Case of *cooperation* (right panel). Example : **env** = 9331, **bs** = 2, **gid** = 1, **factor** = 1.291. The community provides a **gain of about +29%** for the final biomass of the target. The top- ΔS is dominated by **EX_cpd00098_e0 [EX_cpd00098_e0]** with a marked positive change, accompanied by more modest adjustments on other compounds (**EX_cpd00009_e0 [EX_cpd00009_e0]**, **EX_cpd01262_e0 [EX_cpd01262_e0]**, ...). Two compatible mechanisms : (i) **cross-feeding** — a metabolite useful to the target is *produced* by the partner then *used* by the target (lifting a limitation); (ii) **detoxification/removal** — although inhibition is not explicitly modeled in our dFBA, a plausible interpretation is that a partner removes inhibitory compounds, alleviating potential constraints on the target.

Reading guide.

- $\Delta S < 0$ (bar to the left) $\Rightarrow$ **net consumption** : if the same resource is strongly consumed when adding the partner, *co-consumption* and *competition* are likely.
- $\Delta S > 0$ (bar to the right) $\Rightarrow$ **production/accumulation** : useful if the target can use it (cross-feeding), deleterious if the product is unusable or toxic.
- The **factor** aggregates the effect on the target : < 1 (*competition dominant*); > 1 (*contextual cooperation*).

Takeaway. The figure illustrates two archetypes : (i) **co-consumption of a key substrate** or **accumulation of by-products** $\Rightarrow$ **factor** $\ll 1$; (ii) **metabolic supply** (production of a limiting factor, removal of an inhibitor) $\Rightarrow$ **factor** > 1 . This joint reading of *metabolite trajectories* + *final biomass* enables formulation of mechanistic hypotheses to be tested in the *wet lab*.

4.4 Contrasting cases at $g=2$ (environment 15)

For environment 15, we illustrate two co-cultures **target** + **partner**, showing opposite effects on the target's growth : (i) a **beneficial** partner that improves the target's biomass trajectory, and (ii) a **harmful** partner that reduces it. *The target biomass in monoculture (baseline) is not plotted in these figures.*

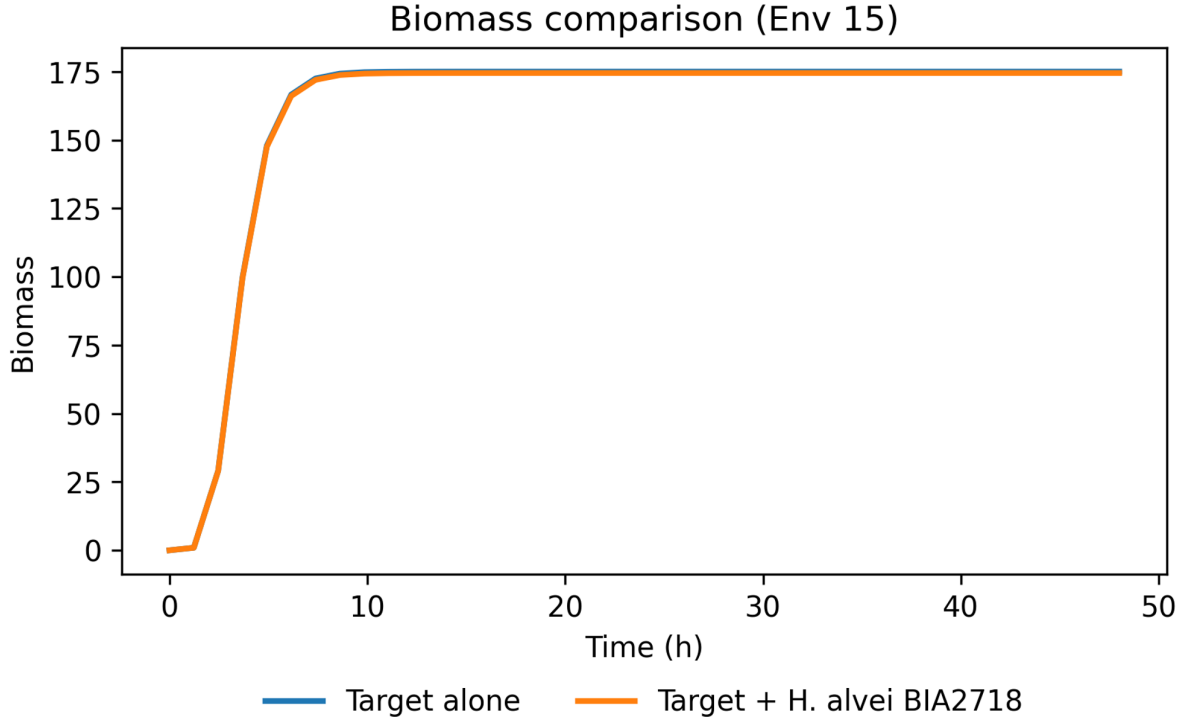


FIGURE 9 – Environment 15, $g=2$ — co-culture **target** + **beneficial partner** : the presence of the partner increases the target's growth. **Curve convention** : *baseline* = target biomass in monoculture; *target* = target biomass *in community*; the other named curves correspond to partner biomasses (members) in the community.

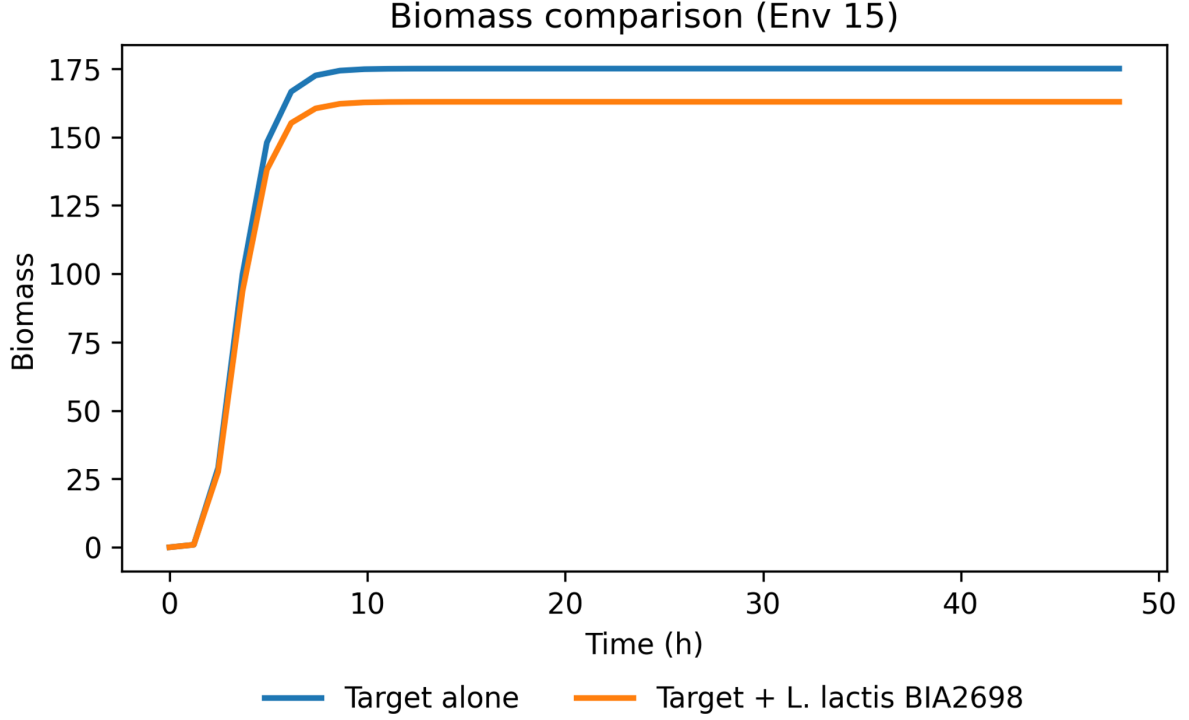


FIGURE 10 – Environment 15, $g=2$ — co-culture **target** + **harmful partner** : the presence of the partner reduces the target’s growth. **Curve convention** : *baseline* = target biomass in monoculture ; *target* = target biomass *in community* ; the other named curves correspond to partner biomasses (members) in the community.

4.5 Change of carbon source & viability controls

Following initial observations (often “*monoculture* > *assembly*”), we tested environments using specific carbon sources (e.g., N-Acetyl-D-glucosamine) and controlled oxygen levels ($O_2 = 0$ or 10). After checking missing pathways (completed manually, ~ 5 reactions), four environments were tested : env_0 (carbon #27 + O_2 0), env_1 (carbon #27 + O_2 10), env_2 (carbon #154 + O_2 0), env_3 (carbon #154 + O_2 10). Figures 11–14 compare, for $g \in \{2, 3, 4, 5\}$, the target’s biomass in contrasted conditions. In these tests, **monoculture** often appears superior, which **reinforces the relevance** of the *in silico* medium retained for the main campaign (10000 media). *Overall, the two newly tested sugars did not provide any advantage*

for collective growth compared to individual growth, under either aerobic or anaerobic conditions.

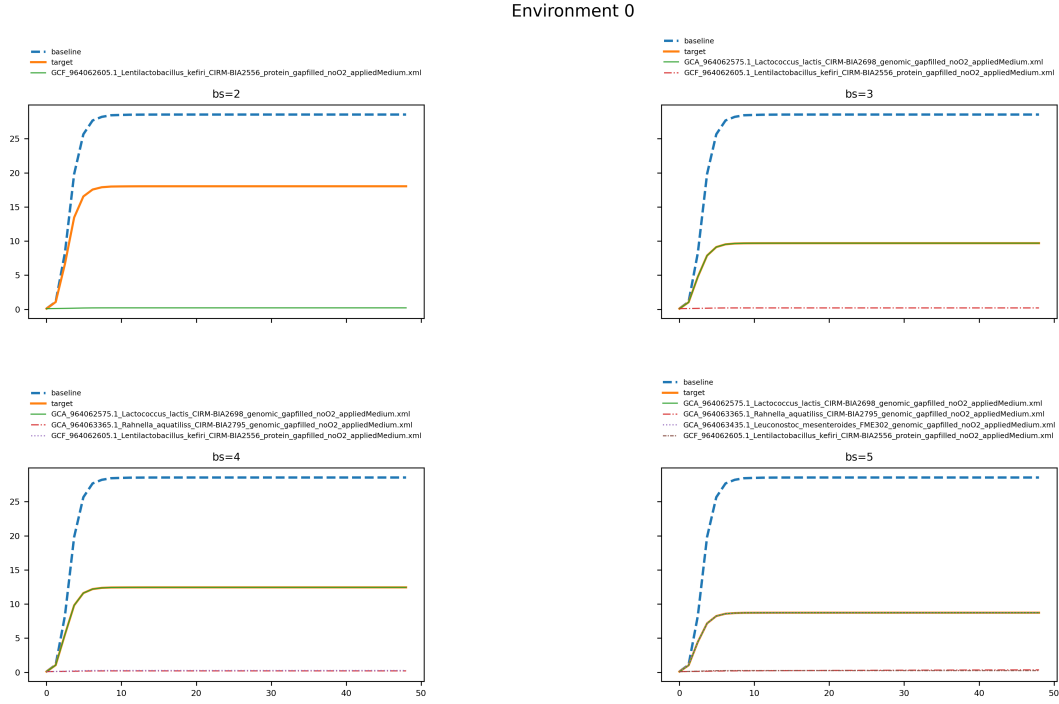


FIGURE 11 – Environment 0 : comparison of target biomasses ($g=2,3,4,5$).
Curve convention : *baseline* = target biomass in monoculture ; *target* = target biomass *in community* ; the other named curves correspond to partner biomasses (members) in the community.

Environment 1

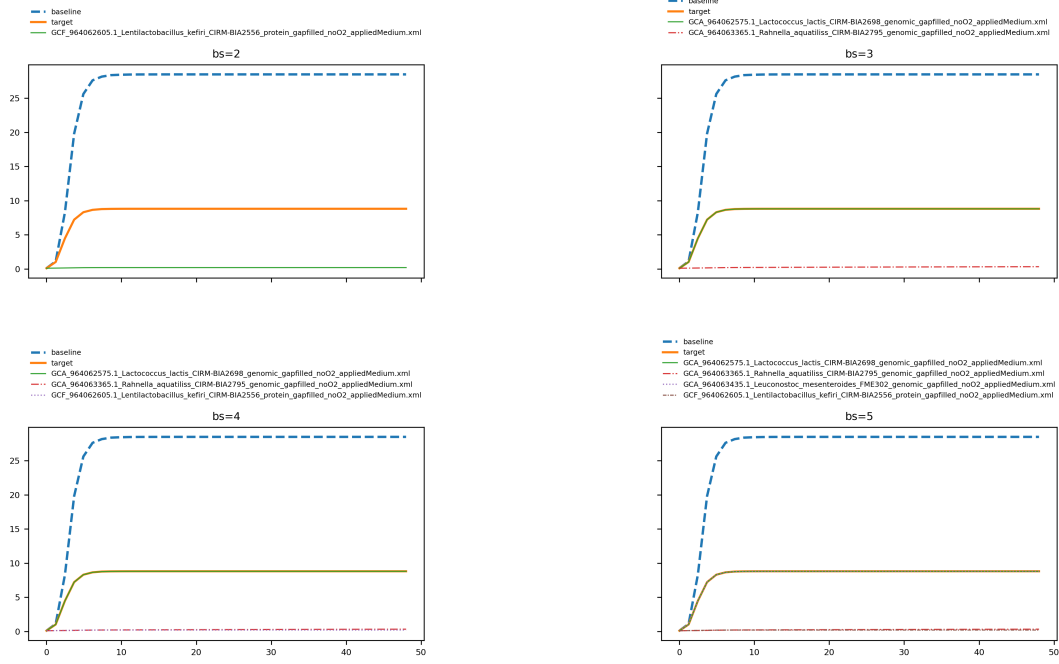


FIGURE 12 – Environment 1 : comparison of target biomasses ($g=2,3,4,5$).
Curve convention : *baseline* = target biomass in monoculture ; *target* = target biomass *in community* ; the other named curves correspond to partner biomasses (members) in the community.

Environment 2

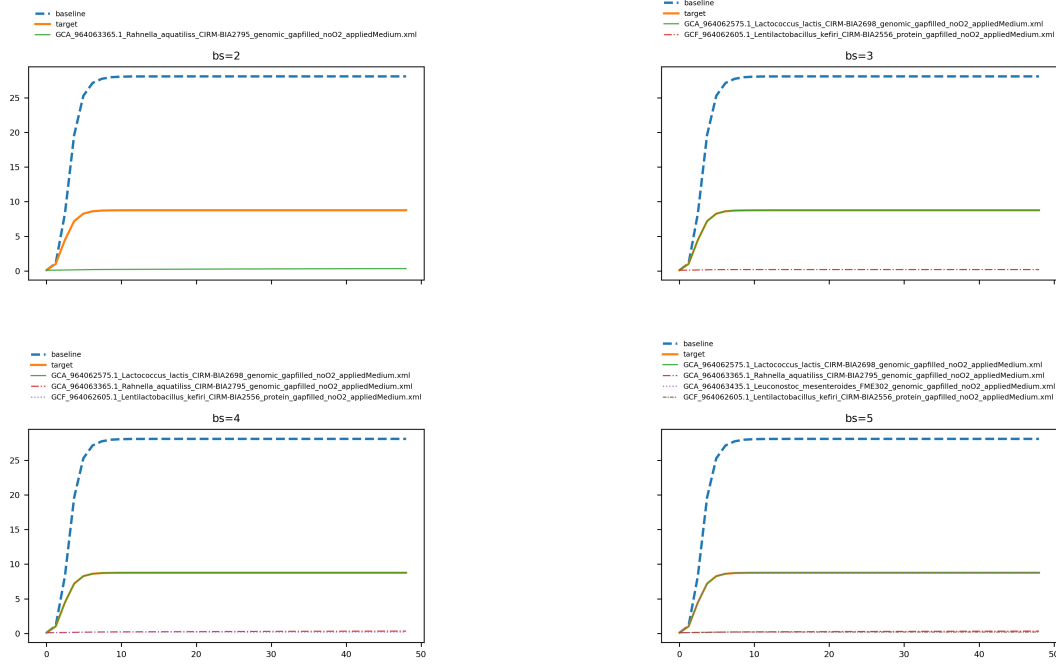


FIGURE 13 – Environment 2 : comparison of target biomasses ($g=2,3,4,5$).
Curve convention : *baseline* = target biomass in monoculture ; *target* = target biomass *in community* ; the other named curves correspond to partner biomasses (members) in the community.

Environment 3

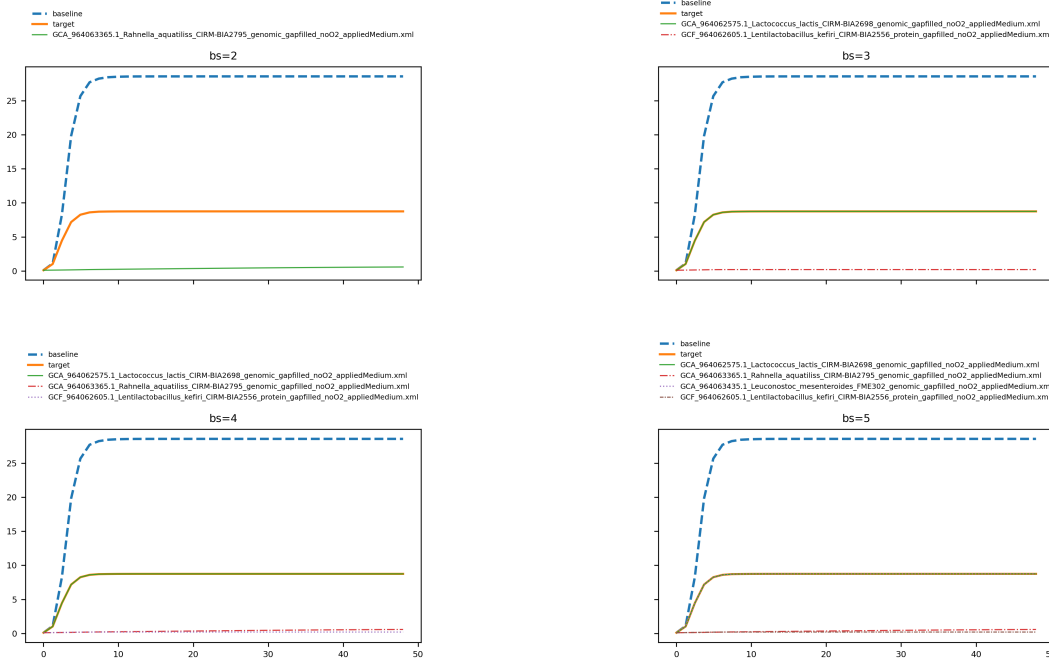


FIGURE 14 – Environment 3 : comparison of target biomasses ($g=2,3,4,5$).
Curve convention : *baseline* = target biomass in monoculture; *target* = target biomass *in community*; the other named curves correspond to partner biomasses (members) in the community.

These four figures show that the two newly tested sugars, when used as carbon sources, do not provide any advantage for collective growth compared to individual growth, under either aerobic or anaerobic conditions.

5 Discussion

Our pipeline highlights two complementary contributions. On the one hand, the **FBA pre-screen + beam search** drastically reduces cost without degrading detection of the best groups, consistent with literature using stationary methods like **SMETANA** or community frameworks for a first screening [14, 15, 12]. On the other hand, **dFBA** reveals transient behaviors (depletion/accumulation, delays in *cross-feeding* appearance, extinctions) inaccessible in stationary analysis [13].

Furthermore, the frequent observation “*monoculture > assembly*” does not contradict the initial objective : **It is a robust observation, already reported in the literature : many communities merely induce nutrient competition** [4]. This reality precisely motivates the fine analysis of media and consumption profiles presented in §4.3, as well as approaches to rank co-cultures among themselves.

We recall the classic limitation of stationary pre-screening : **some higher-order interactions** (multi-species synergies/antagonisms) can be *ignored* by an “instantaneous” FBA. Nevertheless, the best combinations identified by FBA tend to remain competitive once simulated in dFBA, hence the practical value of the scheme “*FBA for preselection* $\rightarrow$ *dFBA for validation*” [14, 15, 12].

Finally, targeted carbon-source trials confirm that medium choice strongly influences the “*monoculture vs assembly*” balance. In the four tested environments (carbons #27/#154 and O₂ 0/10) [Figures 9–12], monoculture often dominates, which **reinforces the relevance** of the *in silico* medium chosen for the main campaign (10 000 media).

6 Conclusion and perspectives

We proposed a *reproducible* pipeline to identify ecological partners favoring colonization of a target strain : homogeneous reconstruction and *gap-filling*, exchange unification, generation of 10 000 media, **FBA pre-screen with beam search** (4 h $\rightarrow$ 9 min/env) then **dFBA** of retained groups, with aggregation of frequencies over all media.

Current state of the data. At this stage, we only have simulation data. Real experimental data (co-cultures, growth curves, metabolite profiles) will be provided by our *wet-lab* collaborators after the August vacation period ; since the defense date occurs before the end of the internship, **the real dataset is not yet available. Without these data**, it is not possible to **build nor validate** a generalized *prediction* model of the best partners.

Plan for the predictive model (as soon as data are available). We propose a **supervised** framework of the *learning-to-rank* type to predict, *per environment*, the partners maximizing the target’s final biomass :

- **Inputs (features) per group** : (i) target μ_{solo} ; (ii) stationary interaction scores derived from FBA/SMETANA (e.g., complementarity of imports/exports, potential exchanges) [14, 15, 12] ; (iii) overlaps of “consumptions/secretions” sets ; (iv) environment vector (or its principal components) ; (v) structural network metrics (degrees/centralities of activated pathways).
- **Output** : rank (or probability) of being *in the best group* for the given environment.
- **Candidate algorithms** : listwise gradient boosting (optimizing NDCG), ranking random forests, or ordinalized regression.
- **Training/Validation** : *cross-validation* by *environments* (avoid leakage), NDCG@k/AUC metrics, and probability *calibration*.

Recent literature confirms that **descriptors from FBA/SMETANA** capture interaction signals useful for prediction [14, 15, 12] ; dFBA and experimental measurements will refine these labels and quantify uncertainty.

Transfer to experimental. Upon receipt of *wet-lab* data, we will (i) confront dFBA trajectories with experimental curves, (ii) correct models (missing transporters/pathways) and (iii) train the *ranking* model to recommend, *per medium*, the partners to test as a priority.

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